

Customer Behavior Analysis

Turning customer transaction data into actionable business insights

MCA Student | Aspiring Data Analyst · [Python](#) · [SQL](#) · [Power BI](#)

Executive Summary

3,900

Total Customers

Customer purchase records analyzed across demographics, products, and behavior

\$59.76

Avg Purchase Value

Average transaction amount across all customer segments

3.75

Avg Review Rating

Customer satisfaction score out of 5, indicating solid but improvable experience

Business Focus Areas

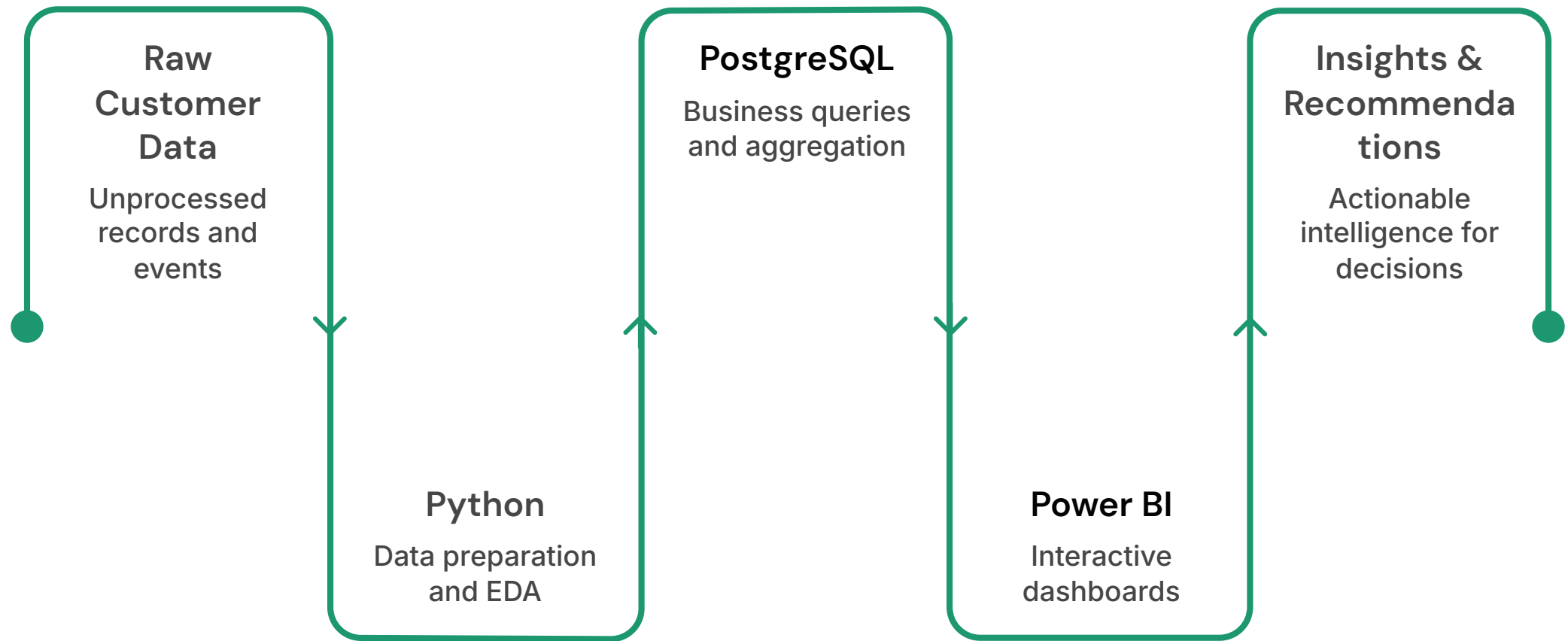
1. Customer Segmentation by purchase frequency and loyalty tier
2. Category Performance across product lines
3. Subscription Behavior and conversion opportunity
4. Purchase Behavior across demographics and shipping preferences

Project Scope

This analysis examines purchasing behavior across **demographics, product categories, subscription status, discount usage, shipping preferences, and purchase frequency**. The goal is to surface patterns that drive retention, revenue, and customer satisfaction — translating raw transactional data into prioritized business recommendations.

From Raw Data to Business Insight

A structured, multi-tool analytical pipeline transforms unprocessed records into decision-ready intelligence.



Why This Stack?

Python handles data cleaning, imputation, and exploratory analysis at scale. **PostgreSQL** enables structured business queries with aggregation and window functions. **Power BI** delivers interactive, stakeholder-ready dashboards. Each tool is chosen for its role in the pipeline — not interchangeably.

Analytical Rigor

The pipeline enforces a clean separation between data preparation, querying, and visualization. This ensures reproducibility, auditability, and a clear chain of logic from raw input to final recommendation — a standard expected in professional analytics engagements.

Data Preparation & Transformation

Dataset Overview

3,900 Records

Customer purchase transactions

18 Columns

Feature dimensions per record

Missing Values

Identified in Review Rating field

Transformations Applied

1 Age Group Segmentation

Continuous age values binned into discrete cohorts to enable demographic analysis and cross-segment comparison.

2 Median Imputation

Missing Review Rating values replaced with the column median — a robust strategy that avoids mean distortion from outliers.

3 Feature Engineering

Derived features constructed to support segmentation, loyalty classification, and behavioral pattern detection.

✓ Result: Clean, structured dataset prepared for PostgreSQL analysis and Power BI visualization.

SQL: Answering Business Questions

10 business questions translated into structured queries — from simple aggregations to advanced window functions.

Core SQL Techniques Used

GROUP BY

Segment-level aggregation across categories, demographics, and shipping types

SUM() / AVG()

Revenue totals and average purchase values per cohort

CASE WHEN

Conditional logic for loyalty tier classification and behavioral flags

ROW_NUMBER()

Window function for within-category product ranking

Advanced Query: Top Products by Category

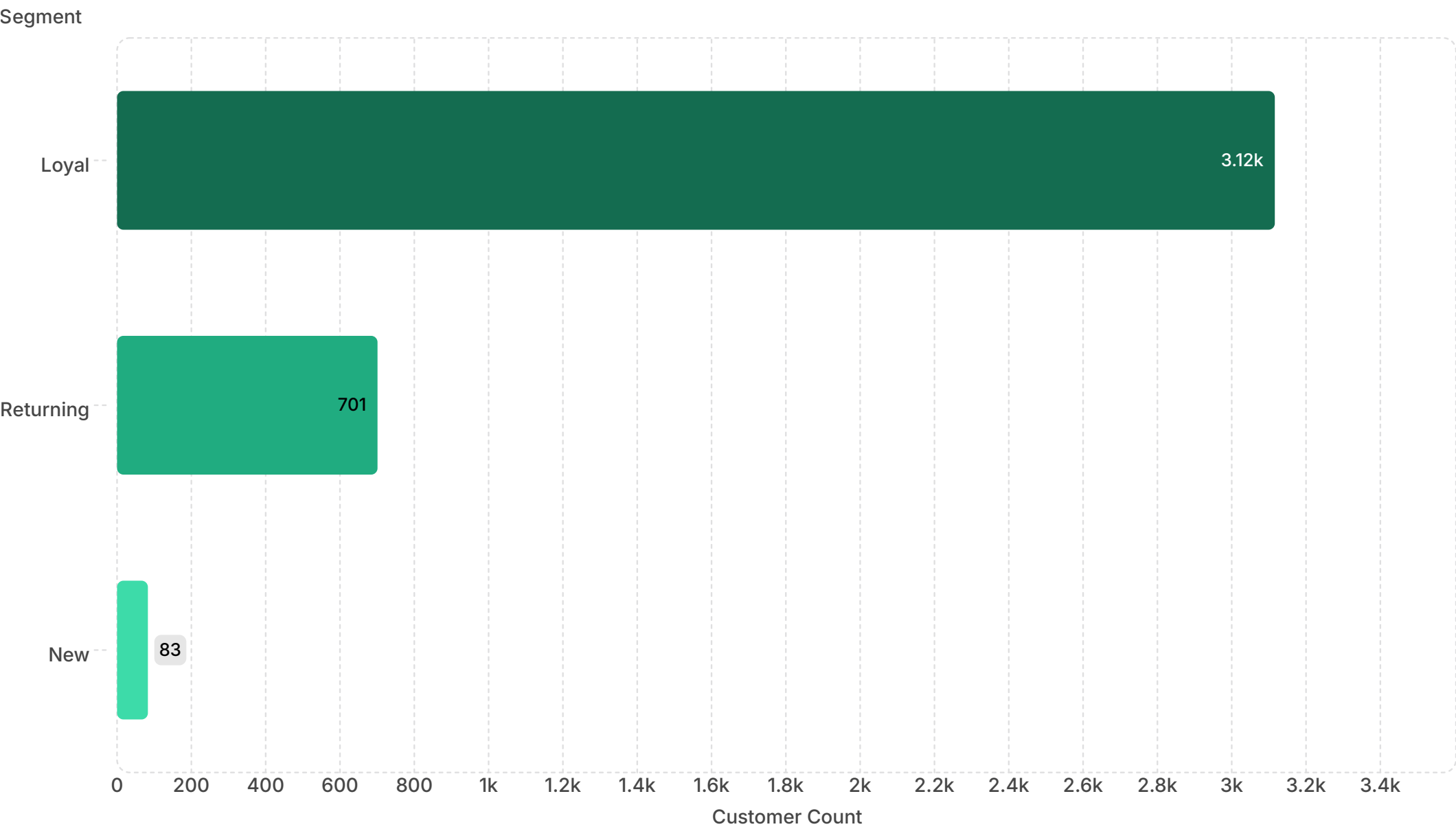
The most analytically sophisticated query in this project uses `ROW_NUMBER()` with `PARTITION BY` category to rank products **independently within each category**. This window function approach avoids the need for multiple subqueries and enables category-specific performance analysis in a single, efficient pass over the data.

```
SELECT category, item, total_revenue,  
       ROW_NUMBER() OVER (  
         PARTITION BY category  
         ORDER BY total_revenue DESC  
       ) AS rank_in_category  
FROM category_revenue  
WHERE rank_in_category = 1;
```

This pattern is directly applicable to inventory prioritization, promotional planning, and category management decisions.

Customer Segmentation

Customers classified into three loyalty tiers based on purchase frequency and recency — revealing a dataset dominated by high-value repeat buyers.



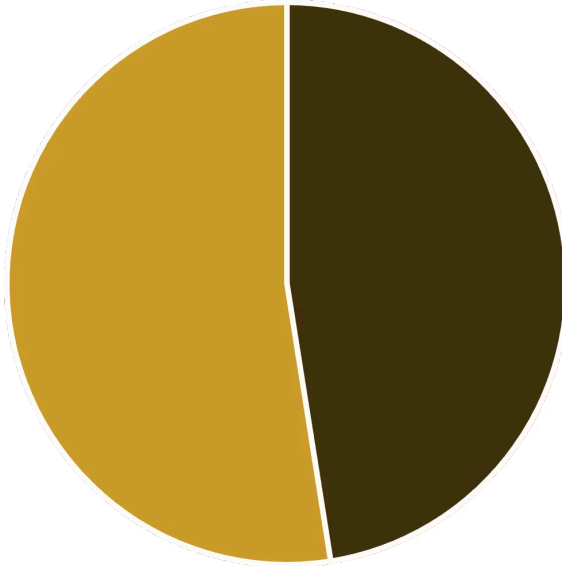
Segment Definitions

Tier	Count	Share	Profile
Loyal	3,116	79.8%	High-value repeat customers
Returning	701	18.0%	Regular shoppers
New	83	2.1%	First-time buyers

Key Finding

Loyal customers represent **79.8%** of the analyzed dataset — an unusually high retention rate indicating strong repeat purchase behavior. The relatively small new customer cohort (83 customers, 2.1%) suggests acquisition channels may be underutilized, while the 701 returning customers represent a critical conversion opportunity into the loyal tier.

Revenue by Gender



Distribution Finding

Female customers generate **slightly higher total revenue** than male customers across the 3,900-record dataset. The split is approximately even — reflecting a balanced customer base rather than a skewed demographic concentration.

Strategic Implications

Gender-Based Marketing

Near-parity in revenue contribution suggests both segments warrant dedicated campaign investment rather than a one-size-fits-all approach.

Product Positioning

Category-level gender breakdowns (not shown here) can inform assortment decisions and targeted promotional strategies.

Personalization Opportunity

Demographic segmentation combined with purchase history enables precision targeting to improve conversion rates across both groups.

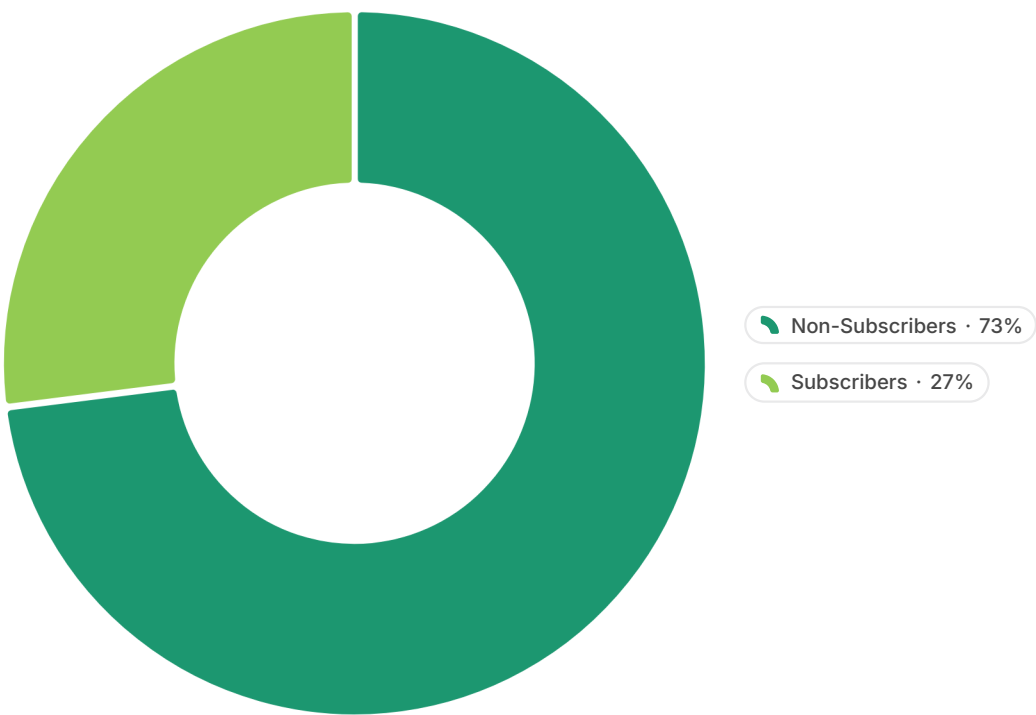
Shipping & Subscription Impact

Shipping Preference Analysis



Express shipping customers spend **12% more** on average (\$65 vs \$58). This correlation suggests express shipping preference is a proxy for higher-intent, higher-value purchasing behavior — a useful signal for upsell and premium tier targeting.

Subscription Status Breakdown



With **73% of customers** currently non-subscribers, the subscription program has significant headroom for growth. Converting even a fraction of this base represents a material recurring revenue opportunity — particularly given the strong loyalty tier already present in the dataset.

Key Insights

Four evidence-based findings drawn directly from the data — each with a clear business implication.

1

Customer Loyalty Dominates

Loyal customers account for **3,116 of 3,900** records (79.8%), confirming strong retention and repeat purchase patterns. Retention infrastructure is working — the priority is now maximizing lifetime value from this cohort.

2

Apparel Categories Lead Revenue

Clothing generates the highest revenue, followed by Accessories. These two categories represent the core of the business and should anchor inventory planning, promotional calendars, and marketing spend allocation.

3

Subscription Gap Is a Growth Lever

The **73% non-subscriber base** is the single largest untapped revenue opportunity identified in this analysis. Targeted conversion campaigns — especially toward the returning customer segment — could materially shift recurring revenue.

4

Satisfaction Has Room to Grow

An average review rating of **3.75 / 5.00** indicates solid but not exceptional customer satisfaction. Identifying the product and experience drivers of lower ratings is a clear path to competitive differentiation.

Strategic Recommendations

Five prioritized actions derived from the analysis — structured for immediate business application.

1 Subscription Growth

Develop targeted subscription offers and loyalty benefits to convert the **73% non-subscriber base**. Prioritize the returning customer segment (701 customers) as the highest-probability conversion cohort given their demonstrated purchase intent.

2 Category Focus

Maintain inventory depth and marketing investment in **Clothing and Accessories** — the top-performing categories by revenue. Use SQL-derived category rankings to inform seasonal assortment and promotional prioritization.

3 Loyalty Program Strengthening

Implement retention programs specifically designed for the **3,116 loyal customers** to maximize lifetime value. Tiered rewards, early access, and personalized offers are high-ROI interventions for this already-engaged cohort.

4 Demographic Targeting

Develop **age-based and gender-based campaigns** using the segmentation features engineered during data preparation. Near-parity in gender revenue contribution supports parallel investment in both segments with differentiated creative and product focus.

5 Quality & Experience Improvement

Investigate products and categories driving ratings below the **3.75 average**. Implement structured feedback loops and product experience improvements to push satisfaction scores above 4.0 — a threshold associated with stronger repeat purchase rates.

📄 This analysis was conducted using Python, PostgreSQL, and Power BI on a dataset of 3,900 customer records. All figures cited are derived directly from the data — no estimates or external benchmarks were applied.